

Problem Set 6

Exercise 1 Consider the IV regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

where X_i is correlated with u_i and Z_i is an instrument. Which IV assumption is not satisfied when

- a) Z_i is independent of (Y_i, X_i, W_i) ?
- b) $Z_i = W_i$
- c) $W_i = 1 \forall i$
- d) $Z_i = X_i$

Exercise 2 A classmate has developed an IV regression model with one regressor, X_i , and two instruments, Z_{1i} and Z_{2i} . She has a strong theoretical basis as to why $\text{corr}(Z_{1i}, u_{i2}) = 0$, namely that Z_{1i} is the result of a random lottery. Preliminary work, however, showed that the first-stage F -statistic from this exactly identified model was insufficiently large for Z_{1i} to be considered a relevant instrument set by itself. As a result she includes an additional instrument, Z_{2i} , which is strongly relevant but is less likely to satisfy the condition of instrument exogeneity. In the instrumental variable regression model with one regressor, X_i , and two instruments, Z_{1i} and Z_{2i} , the value of the J -statistic is $J = 7.5$.

- a) Does this suggest that $\mathbb{E}(u_i | Z_{1i}, Z_{2i}) \neq 0$? Explain.
- b) Does this suggest that $\mathbb{E}(u_i | Z_{2i}) \neq 0$? Explain.

Exercise 3 A researcher is interested in the effect of more secure property rights on income across countries. He collects recent data from 60 countries and runs the OLS regression $Y_i = \beta_0 + \beta_1 X_i + u_i$, where Y_i is a country's GDP per capita and X_i is an index taking values between 0 and 10 reflecting the protection against expropriation where a higher value indicates greater protection against expropriation, that is, more secure property rights.

- a) Explain why the OLS estimates are likely to be unreliable and indicate in which direction they might be biased.
- b) All of the countries in the researcher's sample were former colonies. Institutions securing property rights could originate from early institutions established alongside European settlements. The decision for Europeans to settle or otherwise could reflect concerns for mortality among settlers. Explain how settler mortality might be used as an instrument to estimate the effect of more secure property rights on income across countries.

Exercise 4 A researcher is using a panel data set on $n = 1000$ workers over $T = 10$ years (from 2008 through 2017) that contains the workers' earnings, sex, education, and age. The researcher is interested in the effect of education on earnings.

- a) Give some examples of unobserved person-specific variables that are correlated with both education and earnings.
- b) Can you think of examples of time-specific variables that might be correlated with education and earnings?
- c) How would you control for these person-specific and time-specific effects in a panel data regression?
- d) Can the regression that you suggested in the previous answer be used to estimate the effect of a worker's sex on his or her earnings?
- e) Can that regression be used to estimate the effect of the national unemployment rate on an individual's earnings? Explain.

Exercise 5 Consider the results presented in Table 1.

- a) New Jersey has a population of 8.85 million people. Suppose New Jersey increased the tax on a case of beer by \$2 (in 1988 dollars). Use the results in column (5) to predict the number of lives that would be saved over the next year. Construct a 99% confidence interval for your answer.
- b) The drinking age in New Jersey is 21. Suppose that New Jersey lowered its drinking age to 19. Use the results in column (5) to predict the change in the number of traffic fatalities in the next year. Construct a 95% confidence interval for your answer.
- c) Suppose real income per capita in New Jersey increases by 3% in the next year. Use the results in column (6) to predict the change in the number of traffic fatalities in the next year. Construct a 95% confidence interval for your answer.

Exercise 6 Suppose a researcher believes that the occurrence of natural disasters such as earthquakes leads to increased activity in the construction industry. He decides to collect province-level data on employment in the construction industry of an earthquake-prone country, like Japan, and regress this variable on an indicator variable that equals 1 if an earthquake took place in that province in the last five years.

- a) Should the researcher include province fixed effects in order to control for location-specific characteristics of the labor market?
- b) What can the researcher do to control for location effects?

Exercise 7 a) In the fixed effects regression model, are the fixed entity effects, α_i , consistently estimated as $n \rightarrow \infty$ with T fixed?

- b) If n is large (say, $n = 2000$) but T is small (say, $T = 4$) do you think that the estimated values of α_i are approximately normally distributed? Why or why not?

Dependent variable: traffic fatality rate (deaths per 10,000).

Regressor	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Beer tax	0.36 (0.05) [0.26, 0.46]	-0.66 (0.29) [-1.23, -0.09]	-0.64 (0.36) [-1.35, 0.07]	-0.45 (0.30) [-1.04, 0.14]	-0.69 (0.35) [-1.38, 0.00]	-0.46 (0.31) [-1.07, 0.15]	-0.93 (0.34) [-1.60, -0.26]
Drinking age 18		0.10		0.03 (0.07) [-0.11, 0.17]	-0.01 (0.08) [-0.17, 0.15]		0.04 (0.10) [-0.16, 0.24]
Drinking age 19				-0.02 (0.05) [-0.12, 0.08]	-0.08 (0.07) [-0.21, 0.06]		-0.07 (0.10) [-0.26, 0.13]
Drinking age 20				0.03 (0.05) [-0.07, 0.13]	-0.10 (0.06) [-0.21, 0.01]		-0.11 (0.13) [-0.36, 0.14]
Drinking age						0.00 (0.02) [-0.05, 0.04]	
Mandatory jail or community service?				0.04 (0.10) [-0.17, 0.25]	0.09 (0.11) [-0.14, 0.31]	0.04 (0.10) [-0.17, 0.25]	0.09 (0.16) [-0.24, 0.42]
Average vehicle miles per driver				0.008 (0.007)	0.017 (0.011)	0.009 (0.007)	0.124 (0.049)
Unemployment rate				-0.063 (0.013)		-0.063 (0.013)	-0.091 (0.021)
Real income per capita (logarithm)				1.82 (0.64)		1.79 (0.64)	1.00 (0.68)
Years	1982–88	1982–88	1982–88	1982–88	1982–88	1982–88	1982 & 1988 only
State effects?	no	yes	yes	yes	yes	yes	yes
Time effects?	no	no	yes	yes	yes	yes	yes
Clustered standard errors?	no	yes	yes	yes	yes	yes	yes
F-Statistics and p-Values Testing Exclusion of Groups of Variables							
Time effects = 0			4.22 (0.002)	10.12 (<0.001)	3.48 (0.006)	10.28 (<0.001)	37.49 (<0.001)
Drinking age coefficients = 0				0.35 (0.786)	1.41 (0.253)		0.42 (0.738)
Unemployment rate, income per capita = 0				29.62 (<0.001)		31.96 (<0.001)	25.20 (<0.001)
R^2	0.091	0.889	0.891	0.926	0.893	0.926	0.899

These regressions were estimated using panel data for 48 U.S. states. Regressions (1) through (6) use data for all years 1982 to 1988, and regression (7) uses data from 1982 and 1988 only. The data set is described in Appendix 10.1. Standard errors are given in parentheses under the coefficients, 95% confidence intervals are given in square brackets under the coefficients, and p -values are given in parentheses under the F -statistics.

Table 1

Exercise 8 Let β_1^{DM} denote the entity-demeaned estimator seen in class, and let β_1^{BA} denote the “before and after” estimator without an intercept, so that $\beta_1^{BA} = \frac{\sum_{i=1}^n (X_{i1} - X_{i2})(Y_{i1} - Y_{i2})}{\sum_{i=1}^n (X_{i1} - X_{i2})^2}$. Show that if $T = 2$, $\beta_1^{DM} = \beta_1^{BA}$.

Exercise 9 Suppose a linear probability model yields a predicted value of Y that is equal to 1.3. Explain why this is nonsensical.

Exercise 10 Seven hundred income-earning individuals from a district were randomly selected and asked whether they are government employees ($Gov_i = 1$) or not ($Gov_i = 0$); data were also collected on their gender ($Male_i = 1$ if male and $= 0$ if female) and their years of schooling ($Schooling_i$, in years). Note, *Schooling* refers to the number of years of education received by people ages 25 and older. The following table summarizes several estimated models.

Dependent Variable: Gov

	Probit (1)	Logit (2)	Linear Probability (3)	Probit (4)	Logit (5)	Linear Probability (6)	Probit (7)
Schooling	0.272 (0.029)	0.551 (0.062)	0.035 (0.003)				0.548 (0.091)
Male				-0.242 (0.125)	-0.455 (0.234)	-0.050 (0.025)	4.352 (1.291)
Male $\times$ Schooling							-0.344 (0.096)
Constant	-4.107 (0.358)	-8.146 (0.800)	-0.172 (0.027)	-1.027 (0.098)	-1.717 (0.179)	0.152 (0.021)	-7.702 (1.238)

Table 2

If need be, you can use the [table of the Standard Normal Cumulative Distribution Function](#).

a) Using the results in column (1):

- i – Does the probability of working for the government depend on Schooling? Explain.
- ii – Friedrich Frnrohr has 16 years of schooling. What is the probability that he will be employed by the government?
- iii – Hans Schneider never went to college (12 years of schooling). What is the probability that Hans will get a government job?
- iv – The sample included values of Schooling between 0 and 18 years, and only five people in the sample had more than 15 years of schooling. Günter Mayer has completed his PhD and has been a student for 24 years. What is the model’s prediction for the probability that Günter will be employed by the government? Do you think that this prediction is reliable? Why or why not?

b) Answer the previous questions (a.i-a.iv) using column 2.

- c) *Sketch the predicted probabilities from the probit and logit in columns (1) and (2) for values of Schooling between 0 and 18. Are the probit and logit models similar?*
- d) *Answer the previous questions (a.i-a.iv) using column 3.*
- e) *Sketch the predicted probabilities from the probit and linear probability in columns (1) and (3) as a function of Schooling for values of Schooling between 0 and 18. Do you think that the linear probability is appropriate here? Why or why not?*
- f) *Using the results in columns (4) through (6):*
 - i – *Compute the estimated probability of being employed by the government for men and for women.*
 - ii – *Are the models in (4) through (6) different? Why or why not?*
- g) *Using the results in column (7):*
 - i – *Liam Johansson is a man with 10 years of schooling. What is the probability that government will employ him?*
 - ii – *Anneli Karlsson is a woman with 12 years of schooling. What is the probability that government will employ her?*
 - iii – *Does the effect of schooling on government employment depend on gender? Explain.*